

Technical Note No. 11*
Options, Futures, and Other Derivatives
John Hull

The Manipulation of Credit Transition Matrices

Suppose that $\mathbf{A}$ is an $N \times N$ matrix of credit rating changes in one year such as those discussed in the text. The matrix of credit rating changes in m years is $\mathbf{A}^m$. This can be readily calculated using the normal rules for matrix multiplication.

The matrix corresponding to a shorter period than one year, say six months or one month is more difficult to compute. We first use standard routines to calculate eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N$ and the corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_N$. These have the property that

$$\mathbf{A}\mathbf{x}_i = \lambda_i\mathbf{x}_i \quad (1)$$

Define $\mathbf{X}$ as a matrix whose i th column is $\mathbf{x}_i$ and $\mathbf{\Lambda}$ as a diagonal matrix where the i th diagonal element is λ_i . From equation (1)

$$\mathbf{A}\mathbf{X} = \mathbf{X}\mathbf{\Lambda}$$

so that

$$\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$$

From this it is easy to see that the n th root of A is

$$\mathbf{X}\mathbf{\Lambda}^*\mathbf{X}^{-1}$$

where $\mathbf{\Lambda}^*$ is a diagonal matrix where the i th diagonal element is $\lambda_i^{1/n}$.

Some authors such as Jarrow, Lando, and Turnbull prefer to handle this problem in terms of what is termed a *generator matrix*.¹ This is a matrix $\mathbf{\Gamma}$ such that the transition matrix for a short period of time Δt is $\mathbf{1} + \mathbf{\Gamma}\Delta t$ and the transition matrix for longer period of time, t , is

$$\exp(t\mathbf{\Gamma}) = \sum_{k=0}^{\infty} \frac{(t\mathbf{\Gamma})^k}{k!}$$

* ©Copyright John Hull. All Rights Reserved. This note may be reproduced for use in conjunction with Options, Futures, and Other Derivatives by John C. Hull.

¹ See R. A. Jarrow, D. Lando, and S.M. Turnbull, "A Markov model for the term structure of credit spreads" *Review of Financial Studies*, 10 (1997), 481–523.